Fleet condition report: Synthetic ACME-style vendor export generated from Kelmarsh KEL01-02

Synthetic ACME-style vendor export generated from Kelmarsh KEL01-02, 2022. 2 turbines, SCADA 01 Jan 2022 to 31 Dec 2022; assessment window: the last 24 months of each turbine (Jan 2022 to Dec 2022). Generated by `windml report` from the mapping `acme_messy.yaml`.

Executive summary

All 2 turbines are green: no validated warning sign in the data up to 31 Dec 2022. Green is not a guarantee; see the notes and limits below.

Green: KEL01, KEL02.

How the ratings work. A turbine is amber only for a specific, unusual pattern: gear-oil pump pressure building up far beyond its sister turbines, repeated oil-system or pitch trips, or a faulty sensor. It is red when the pattern is strong (e.g. a pump pressure rise of 2 bar or more) or persistent and still active. Thresholds were chosen on 2017-2020 data from two UK wind farms and checked on 2021-2024: amber or red turbine-months were followed by a logged fault, trip or long service stop within 90 days 76% of the time, against 50% for random turbines in the same months; but only 19% of the turbine-months before such an event carried a flag. Ratings help decide where to look first; most faults still arrive without a warning sign in SCADA data. With only 2 turbines there is no peer comparison, so only very large changes can be flagged.

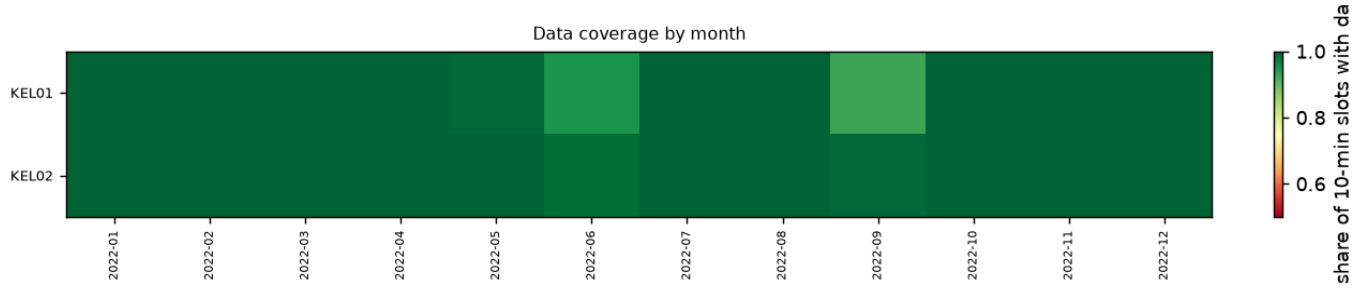
How sure are we. Confidence is *High* when the turbine has 3+ years of history, near-complete data, models trained on earlier years, an alarm log and at least 3 sister turbines to compare with; it drops when any of these is missing. A red rating is a reason to look, not a diagnosis.

What this report could not check:

- The history is short (under two years), so models were cross-fitted within the record: they spot sudden changes and outliers, but a slow change that started before the data begins is invisible.
- The export has no pitch-motor temperatures, so those components could not be checked.

1. Data quality

Turbine	First data	Last data	Months	Coverage (window)	Longest gap (h)	Signals	Native interval	Frozen rows (window)	Stuck sensors (>= 1%)	Stop/fault labels
KEL01	2022-01-01	2022-12-31	12	99.0%	49	6	10min	0	-	alarm log
KEL02	2022-01-01	2022-12-31	12	99.9%	5	6	10min	0	-	alarm log



Signals present in at least one turbine: `ambient_temp_c`, `gear_oil_inlet_pressure_bar`, `gear_oil_temp_c`, `generator_speed_rpm`, `main_bearing_front_temp_c`, `nacelle_position_deg`, `nacelle_temp_c`, `pitch_a_deg`, `power_kw`, `rotor_speed_rpm`, `wind_direction_deg`, `wind_speed_ms`.

2. Turbine scorecard

Turbine	Flag	Confidence	Tech. avail.	Energy avail.	Fault h/yr	Largest fault cause	Oil ep.	Pitch ep.	Abnormal oil days, final yr	Under-perf. days, final yr	Pump pressure: base to last (bar)	Oil-temp model error
KEL01	GREEN	Low	98.5%	-	17	hydraulics (8 h)	0	1	2%	0%	-	-
KEL02	GREEN	Low	97.3%	-	12	converter (6 h)	1	0	2%	0%	-	-

Ratings, reasons and notes

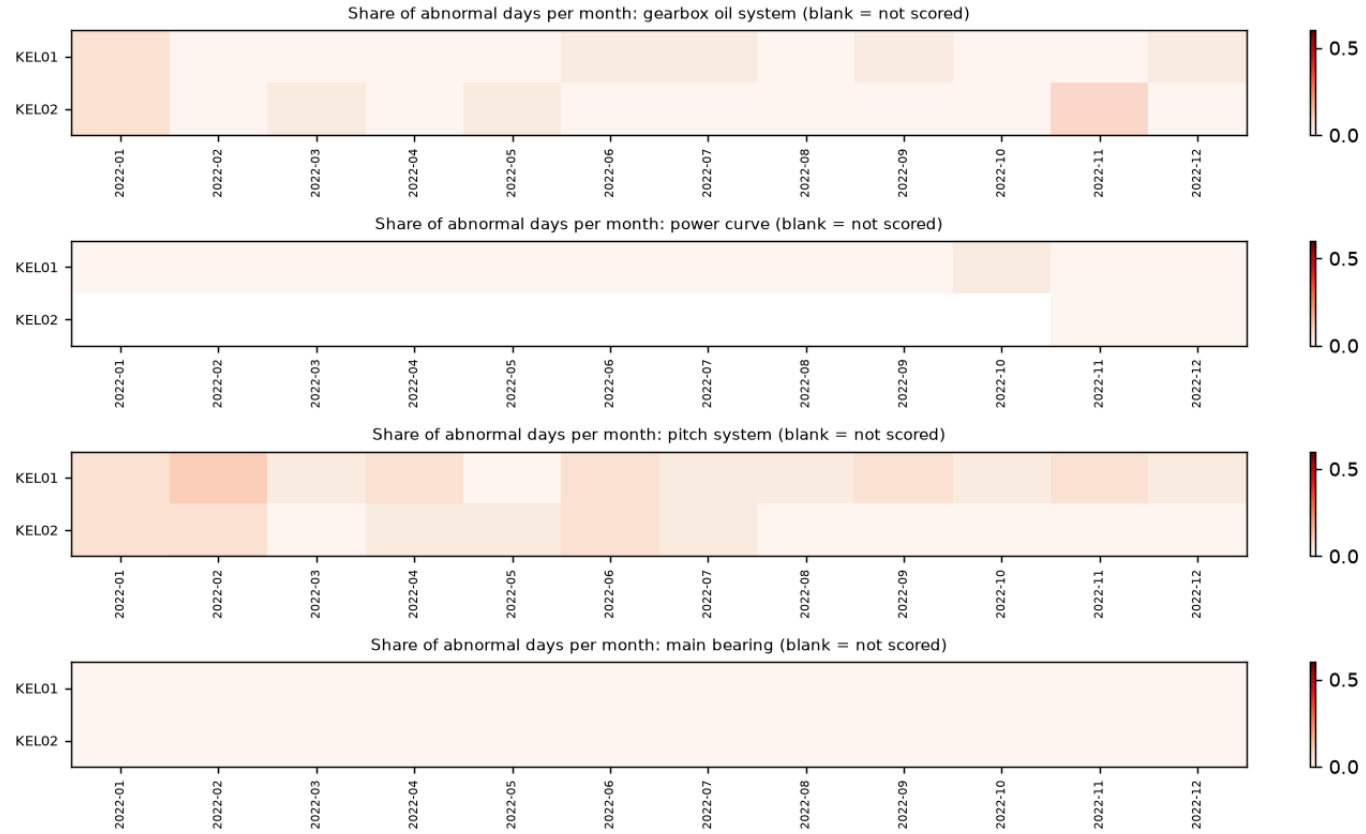
- **KEL01** GREEN (confidence Low: 12 months of history, short-history models (cross-fit), only 2 turbines (no peer comparison)). No rating trigger.
- **KEL02** GREEN (confidence Low: 12 months of history, short-history models (cross-fit), only 2 turbines (no peer comparison)). No rating trigger.

3. Model-based health (normal-behaviour residuals)

Each turbine gets its own models that predict a signal (oil temperature, bearing temperature, pitch angle, power...) from operating conditions. A day is *abnormal* when the turbine's median deviation from its own model is beyond 3 robust standard deviations in the harmful direction. *Chronic* means at least 30% abnormal days in a month.

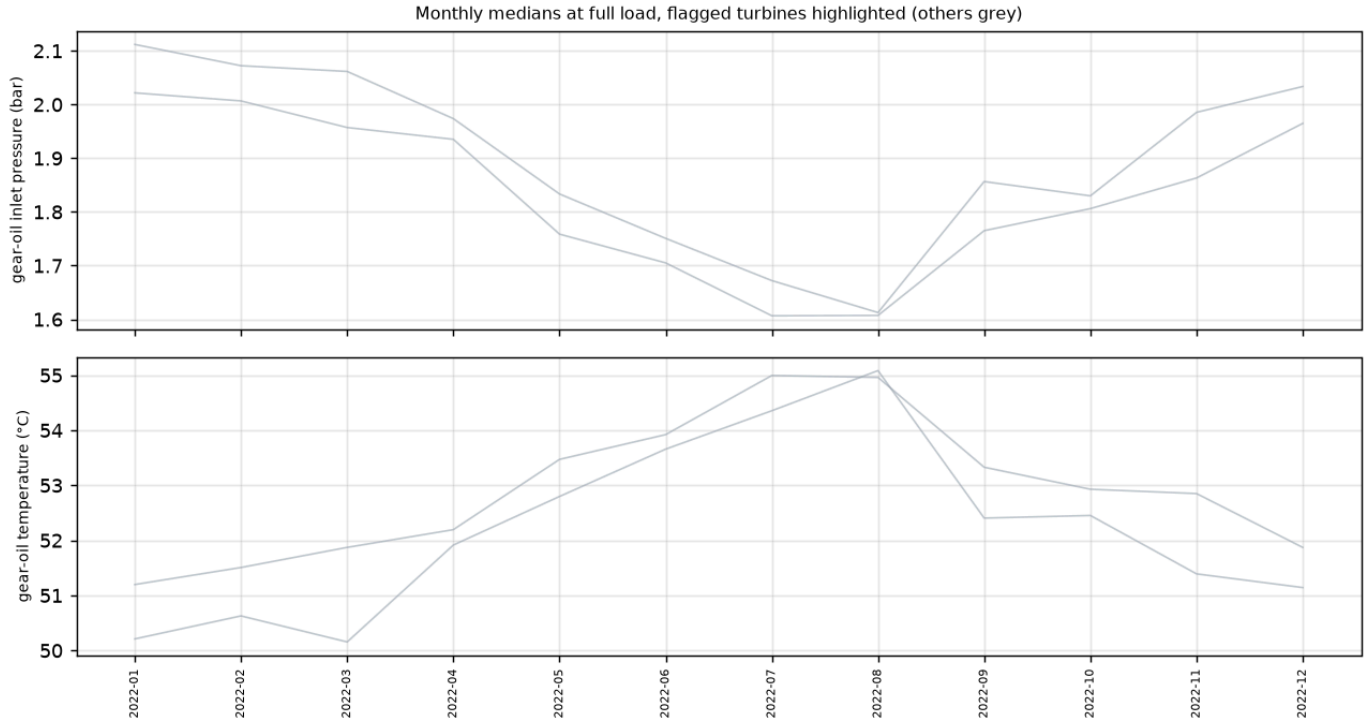
Turbine	Training scheme	Models	Window scored	gearbox oil system: abnormal days final yr / chronic months	main bearing: abnormal days final yr / chronic months	pitch system: abnormal days final yr / chronic months	power curve: abnormal days final yr / chronic months
KEL01	cross-fit	5	99%	2% / 0	0% / 0	5% / 0	0% / 0
KEL02	cross-fit	5	97%	2% / 0	0% / 0	3% / 0	0% / 0

No turbine has 3 or more consecutive chronic months in the window.



4. Gearbox oil pressure and temperature trends

Turbine	Pump pressure (bar): baseline → last month	Inlet pressure (bar)	Oil temperature (°C)	Baseline months
KEL01	-	2.0 → 2.0 (+0.0)	51.3 → 51.1 (-0.1)	2022-01..2022-06
KEL02	-	1.9 → 2.0 (+0.0)	52.0 → 51.9 (-0.2)	2022-01..2022-06

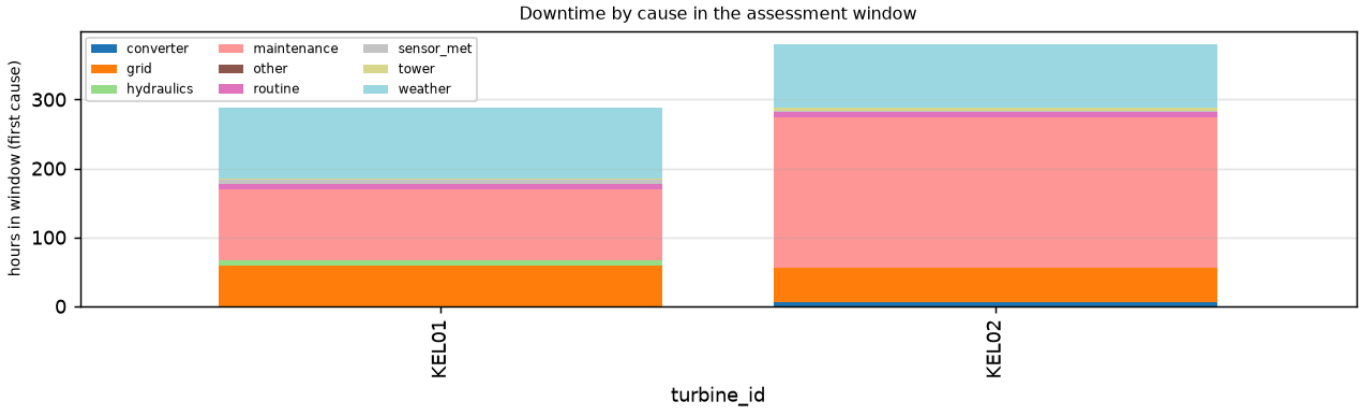


5. Sensor drift and stuck sensors

Fewer than 4 turbines share a drift-check signal, so sensor drift against the fleet could not be checked.

6. Stop and fault history

Turbine	Fault stops	Maintenance	Grid	Weather	Curtailment	Major-fault episodes	Maint. h/yr	Oil / pitch trips
KEL01	75	36	4	12	0	1	102	0 / 1
KEL02	35	30	18	3	0	1	218	1 / 0



Most frequent fault stops in the window (all turbines):

Code	Message	Component	Stops
6530	Anemometer defect	sensor_met	70
4540	Tower oscillation X level 2	tower	12
1070	Drive train monitor level 2	gearbox	6
4510	Tower oscillation Y level 1	tower	6
6620	Vane defect	sensor_met	4
3110	Frequency converter error	converter	4
5510	Low hydraulic pressure	hydraulics	2
4520	Tower oscillation X level 1	tower	2
1065	Drive train monitor level 1	gearbox	1
815	Rotor overspeed nacelle	rotor	1
455	Repeating error BP52	pitch	1
1800	Overload gear oil pump	gearbox	1

7. Methods, thresholds and limits

- **Pipeline.** `windml report` reads the raw export only through the mapping (`windml.ingest`), cleans and labels each turbine (QC flags, short-gap interpolation, alarm-log stop labels), trains the models, computes the metrics and writes this report. Every number here is reproducible from the mapping and the raw files.
- **Window.** The last 24 months of each turbine; *final year* = its last 12 months.
- **Models.** LightGBM per turbine and signal on healthy operating rows (no stop within 6 h, no fault, not in the 30 days before a major fault, QC clean). With 2+ calendar years, each year is scored by a model trained only on earlier years (3-day embargo, training must cover all 12 months). With 12-23 months, the record is cut into 6 blocks scored by models trained on the other blocks (cross-fit). Under 12 months, no model.
- **Ratings.** Computed at each month-end from data up to that date (latest month-end shown here). AMBER when: the full-load gear-oil pump pressure (median of the last 15 days) is >= 1 bar above the turbine's own baseline (days 60-365 before) AND >= 5 robust standard deviations above the other turbines' rise that month; >= 3 oil-system trip episodes in 6 months; >= 6 pitch trip episodes in 6 months; an ambient-temperature or wind-speed sensor drifting vs the fleet over the last 3 months (1.5 °C, 5%); or a monitored sensor frozen on more than 2% of the last 90 days. RED when the pump rise is >= 2 bar; or a trip rule has held 2 month-ends in a row with >= 3x the amber count and a trip in the last 30 days; or two independent systems are amber at once. Shown as notes but never colour-setting, because they had no reliable lead on events in the validation: low availability (it describes the past), model-based abnormal-day rates (oil, pitch, main bearing, power curve) and missing data. A trip episode merges trips less than 24 h apart.
- **Validation.** Retrospective check on Kelmarsh and Penmanshiel (21 turbines). Thresholds selected on 2017-2020, checked on 2021-2024 (792 turbine-months): pump build-up → oil trip or service stop: 58% of 12 flagged turbine-months vs 55% by chance; repeat oil trips → further oil trips: 63% of 75 flagged turbine-months vs 32% by chance; repeat pitch trips → further pitch trips: 67% of 6 flagged turbine-months vs 36% by chance. Any amber or red: 76% followed by a logged event within 90 days vs 50% by chance (recall 19%); red alone 65% vs 58% on only 17 turbine-months, so red marks strength and persistence more than a higher probability. Held-out ratings: 89% green, 9% amber, 2% red. Consecutive months overlap, so the effective sample is much smaller than the row counts; details in `reports/fleet_proof/calibration.md`.
- **Confidence.** Points for history (3+ years: 2, 2-3 years: 1), coverage >= 95%, models trained on earlier years, an alarm log, and a fleet of 4+ turbines (peer comparison). High >= 5, Medium 3-4, Low <= 2.
- **Limits.** The alarm-log classes come from the mapping's rules; trip detection uses Senvion-style oil/pitch codes and messages (other vendors need `class_rules`). Full-load oil medians assume one oil circuit design across the fleet. Models learn whatever was normal in their training data, including a fault that was already present. Sensor drift is relative to the other turbines, so a fleet-wide drift is invisible. The rating thresholds were calibrated on two UK farms of one turbine family (Senvion MM92/MM82); on other fleets the peer-relative logic still applies but the hit rates are unknown. Ratings look back up to 6 months and only at what SCADA and the alarm log show: inspections, oil analyses, vibration (CMS) data and work orders are not used, so a recent repair or a fault the sensors cannot see changes nothing here.
- **Data.** Derived from Kelmarsh SCADA, CC BY 4.0, Cubico Sustainable Investments, doi:10.5281/zenodo.16807551.